



Misalignment-eliminated warping image stitching method with grid-based motion statistics matching

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Abstract

Aligning images is one of the main goals of image stitching. Limited by matching accuracy and deformability, the results of current mainstream stitching approaches in large parallax scenes usually contain obvious stitching errors. A misalignment-eliminated warping image stitching based on grid-based motion statistics (GMS) matching is proposed. A matching approach from coarse-to-fine composed of oriented FAST and rotated BRIEF (ORB) and GMS is integrated to provide more accurate and higher number of inliers for subsequent warping. The local homography with global similarity transformation constraint is used to warp the images to achieve initial image alignment. For the projection biases after local warping, a post-processing step based on thin plate spline (TPS) is proposed for further correction. Both qualitative and quantitative comparisons in experiments of challenging cases show that this method can accurately align images while maintaining a natural look at the same time.

Keywords Image stitching · Image alignment · Local warping · Image registration

1 Introduction

Due to the limited field of view and the fixed resolution of a single lens, it is difficult to balance the high resolution and wide field of view of images in a large viewing angle scene. Therefore, the image stitching algorithm has become the mainstream wide-field image acquisition method, which is widely used in panoramic photography [13], virtual reality [1], and video surveillance [18], and there are bright application prospects in emerging fields such as

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autonomous driving [5] and virtual reality [1]. Given a pair of images with overlapped area, image stitching focuses on estimating the precise warping relationship between them based on mapping correspondence. Previous stitching methods focused on getting the global homography transformation between images, one of the representative methods is Autostitch [4]. Global method treats the deformation between images as the transformation from 2D plane to another, it usually perform well under ideal conditions. However, due to parallax caused by non-ideal conditions such as different lens parameters and shooting angles, there may be multiple degrees of depth in the image domain, resulting in serious ghosting in the results [29]. Ensuring the high quality of the stitching results under large parallax scenes is still one of current main challenges.

Spatially-varying warping models were introduced in some approaches for more accurate alignment in recent years [12, 21, 32]. The representative method is as-projective-as-possible (APAP) [32], which calculated multiple local homography for warping to make images more aligned. But it can't deal with the perspective distortion in non-overlapping areas. The global similarity transformation was introduced in some approaches to make the stitching results look more natural [7, 8, 23]. Some methods achieved higher stitching quality by finding the optimal seamline [6, 17]. However, since there is no further alignment strategy, the actual improvement is limited. Additionally, researchers have tried to apply neural networks to image stitching in the past two years [26]. But learning-based methods are not mature enough, they only perform well on synthetic images without parallax.

These spatially-varying warping approaches are the mainstream of parallax image stitching in recent years, and they usually have good local adaptability under parallax cases. However, due to unreliable feature correspondences and limited deformability, sometimes projection biases may still occur after local warping, resulting in misalignments. The property of these warping methods depends to a great extent on the quantity and quality of the corresponding features [32]. Almost all of these above-mentioned approaches used scale-invariant feature transformation (SIFT) [25] and random sample consensus (RANSAC) [11] for feature extraction and mismatch-elimination, respectively. This combination may easily lead to insufficient or incorrect matches in complex application scenes (such as repetitively-textured areas or multiple planes), resulting in obvious ghosting [19]. The grid-based motion statistics (GMS) matching algorithm is proposed by Bian et al. [2]. GMS converted the motion smoothness constraints into statistical measures to eliminate false matches, that is, more feature point matches are converted into higher quality inliers, which is consistent with our goal of seeking a large amount of accurate feature correspondences.

All these warping methods mentioned above regard stitching as the construction of a geometric transformation model, which limits the deformation effect in areas where the depth of field changes dramatically. Moreover, few studies have focused on post-processing to further eliminate these projection biases after image warping. Thin plate spline (TPS), which reflects the shape of thin metal sheets with physical bending, is a common non-rigid deformation method in the field of medical imaging registration [27]. With the flexible deformability of TPS, the elimination of misalignments is regarded as discrete data approximation processing, so that the projection biases can be accurately described and effectively corrected. But achieving high property of TPS also requires higher quality matching data for support, which also places higher requirements on image matching methods.

The above analysis shows that a stitching method that can obtain high-quality feature correspondences and have strong alignment capabilities in parallax scenes is needed. In this paper, an image stitching method based on GMS and TPS is proposed to further eliminate the

misalignment caused by the projection biases. This method achieved higher alignment accuracy and robustness through two improvements: First, a registration strategy consisting of ORB and GMS is integrated to extract more precise matching correspondences. Second, a warping post-processing step based on TPS is proposed to further optimize the image alignment. Additionally, a global similarity constraint strategy is combined in local warping to correct the perspective distortions. The experimental results on the public stitched image dataset show that compared with other representative methods, although the proposed method has no advantage in computational efficiency, it can achieve much higher alignment accuracy to obtain seamless stitching results while maintaining a natural look.

The rest of this paper is organized as follows. Section 2 reviews the related works of image registration and warping in image stitching. Section 3 describes the proposed method in detail. Experimental results and analyses are presented in Section 4. The conclusion and future work are drawn in Section 5.

2 Related work

2.1 Image registration

Image registration provides important data support for subsequent work in image stitching, which is directly related to the quality of the results [9]. At present, feature-based methods are the mainstream of registration strategies, which is generally combined with two steps: feature extraction and feature matching. For feature extraction, corner detection was the main focus in the early days, such as Harris corner algorithm [16] and features from accelerated segment test (FAST) [10] corner algorithm. Since the corner points contain limited information, Lowe proposed the SIFT descriptor [25], which used the Gaussian difference operator to construct a 128-dimensional feature point descriptor. But SIFT feature descriptors are sparse and may be insufficient in weakly-textured or repetitively-textured areas [19]. Binary descriptors have the advantages of efficient bitwise calculation and high compactness, and usually perform better than corner point and traditional descriptors in image registration. For example, Wang et al. proposed the ORB algorithm [28], which can extract more and denser feature points in a shorter time than SIFT.

In recent years, researchers have searched for new feature description methods to achieve higher-quality registration. Han et al. used hybrid visual features (lines and interest points) to improve the registration accuracy of visible and infrared images [14]. Wu et al. proposed an unsupervised deep binary descriptor (UDBD) [30] based on deep learning, which solved the limitation of handmade binary descriptor such as vulnerable to the geometric transformations and incapable of preserving the manifold structure when learning binary codes.

After feature extraction, feature matching step usually builds the precise correspondences between two image point sets through coarse and fine matching. Coarse matching only performs simple matching through the similarity between feature descriptors, such as the brute force (BF) algorithm [22]. The result of coarse matching usually contains a large number of mismatches, thus other fine matching methods are needed to complete further mismatches elimination. Most recently approaches used the RANSAC algorithm as an advanced feature matching method in image stitching. In fact, RANSAC is essentially a least square method, which sometimes inevitably removes some correct inliers but retains some outliers in multiple homography scenes [33]. Bian et al. proposed a GMS algorithm [2], which removed false

matches by counting the number of features with matching relationships in regions adjacent to coarse matches, and it is not restricted by multiple homography. In other words, GMS converted more matches into higher matching accuracy. Therefore, the coarse-to-fine matching method composed of ORB and GMS can achieve more ideal results in image stitching.

2.2 Spatially-varying warping

Since the global homography method cannot handle non-coplanar scenes, some methods used multiple homography matrices to warp images, collectively called spatially-varying warping methods. Gao et al. made a simple extension on global homography and proposed a dual-homography warping (DHW) method [12], which calculated the final homography by dividing the image into two planes. But DHW can't handle multi-plane situations. APAP warping proposed by Zaragoza et al. [32] divided the images into grid meshes and improved the alignment quality by calculating the local homography of each grid cell. Zhang proposed a multi-view panoramic stitching method for wide baseline images based on local DLT [33]. In order to provide more matching information, Li S et al. used line features as a supplement to point features and achieved good results [19]. But the high computational complexity leads to limited use range.

Several methods focus on correcting the perspective distortions in non-overlapping region. Shape preserving half-projective warping (SPHP) [7] was proposed by Chang. With reference to the shape-preserving method of image scaling, it realized the gradual change across the image from projective to similarity transformation. Adaptive as-natural-as-possible(AANAP) warping proposed by Lin et al. selected the optimal similarity transformation to combine with local homography to reduce perspective distortion [23]. However, they can only deal with the parallax scene with continuous changes in depth of field, it is difficult to guarantee the stitching quality in more complex situations.

2.3 Thin plate spline

Different from traditional image registration, medical image registration [34, 35] is oriented to human organs and tissues that are easily deformed, so non-rigid deformation is the focus of research in this field. Among them, thin plate spline (TPS) is an effective and practical method for approximating large scattered data [3, 15]. A specific linear system is constructed through the correspondence of feature points, and the images are aligned using non-rigid deformation after solving the relevant parameters. Li [20] proposed a robust elastic warping method that used TPS to address parallax problems. However, it has not received much research attention so far. Moreover, it needs to match high-quality data to achieve high deformability.

3 Methods

Parallax processing has always been one of the main challenges in image stitching, and multiple local warping is the current mainstream solution. APAP [32] used moving direct linear transformation (m-DLT) to calculate local transformations through weighted matching data and has made a breakthrough in aligning images, but its alignment capability was affected by limited matching accuracy and deformability. In this paper, the local warping approach in APAP is used as the basis, which is described in detail in section 3.1.

3.1 Local homography warping based on m-DLT

Let I_a and I_b be named target image and reference image, respectively. $\mathbf{p} = [x, y]^T$ and $\mathbf{q} = [u, v]^T$ denote matching points between I_a and I_b . $\mathbf{H}$ is the global transformation matrix between I_a and I_b . The relationship between $\mathbf{p}$ and $\mathbf{q}$ is

$$\hat{\mathbf{q}} = \mathbf{H}\hat{\mathbf{p}}, \quad (1)$$

where $\hat{\mathbf{p}}$ and $\hat{\mathbf{q}}$ is the homogeneous coordinates of $\mathbf{p}$ and $\mathbf{q}$, $\mathbf{H} \in \mathbb{R}^{3 \times 3}$, $\mathbb{R}$ is the set of real number. Let $\mathbf{h}_j^T$ be the j -th row of $\mathbf{H}$. The implicit condition of Eq. (1) is $\mathbf{0}_{3 \times 1} = \hat{\mathbf{q}} \times \mathbf{H}\hat{\mathbf{p}}$, which can be rewritten as follows:

$$\mathbf{0}_{3 \times 1} = \begin{bmatrix} \mathbf{0}_{1 \times 3} & -\hat{\mathbf{p}}^T & v\hat{\mathbf{p}}^T \\ \hat{\mathbf{p}}^T & \mathbf{0}_{1 \times 3} & -u\hat{\mathbf{p}}^T \\ -v\hat{\mathbf{p}}^T & u\hat{\mathbf{p}}^T & \mathbf{0}_{1 \times 3} \end{bmatrix} \mathbf{h}, \mathbf{h} = \begin{bmatrix} \mathbf{h}_1 \\ \mathbf{h}_2 \\ \mathbf{h}_3 \end{bmatrix} \quad (2)$$

For the i -th inlier $\{\mathbf{p}_i, \mathbf{q}_i\}$, let $\mathbf{a}_i \in \mathbb{R}^{2 \times 9}$ be the first-two rows of the matrix in Eq. (2), $i = 1, 2, \dots, N$. $\mathbf{H}$ can be estimated as follows:

$$\mathbf{h} = \text{hargmin} \sum_{i=1}^N \|\mathbf{a}_i \mathbf{h}\|^2 = \text{hargmin} \|\mathbf{A} \mathbf{h}\|^2, \text{ s.t. } \|\mathbf{h}\| = 1 \quad (3)$$

where $\mathbf{A} \in \mathbb{R}^{2N \times 9}$ is obtained by arranging $\mathbf{a}_i$ in order of i . The global homography matrix $\mathbf{H}$ is reshaped from $\mathbf{h}$.

Since practical stitching scenes usually contain multiple planes, only using global homography warping cannot align the image and will lead to obvious ghosting. In [32], the input image is divided into $C1 \times C2$ cells, and m-DLT is used to estimate local homography warping at each grid point to achieve more accurate alignment.

Let $\mathbf{p}_j$ be the coordinate of grid point, $\mathbf{H}_j$ is the local homography at $\mathbf{p}_j$. Through m-DLT, $\mathbf{H}_j$ is estimated as follows:

$$\mathbf{h}_j = \text{hargmin} \sum_{i=1}^N \left\| \omega_j^i \mathbf{a}_i \mathbf{h} \right\|^2 = \text{hargmin} \left\| \mathbf{W}_j \mathbf{A} \mathbf{h} \right\|^2 \quad (4)$$

$\{\omega_j^i\}_{i=1}^N$ are scalar weights which are estimated as $\omega_j^i = \max\left(\exp\left(-\|\mathbf{p}_j - \mathbf{p}_i\|^2 / \sigma^2\right), \gamma\right)$, which means that ω_j^i changes according to $\mathbf{p}_j$. $\mathbf{p}_i$ is one point the i -th match pair $\{\mathbf{p}^i, \mathbf{q}^i\}$, σ is a scale parameter, $\gamma \in [0, 1]$ is an offset, which prevents numerical problems in estimation. $\mathbf{W}_j$ is a weight matrix which is composed as $\mathbf{W}_j = \text{diag}\left(\left[\omega_j^1 \omega_j^1 \omega_j^2 \omega_j^2 \omega_j^N \omega_j^N\right]\right)$. A higher γ can transform the warp of the non-overlapping area into a global projection warp to reduce the extrapolation distortion. Nevertheless, the perspective distortion in non-overlapping areas is inevitable.

3.2 Misalignment-eliminated image stitching method

It can be seen from the Eq. (3) that in theory only 4 pairs of inliers are needed to obtain the global homography matrix. But local warping puts forward higher requirements on the quantity and

accuracy of feature matching. In m-DLT, the value of $\mathbf{h}_j$ depends on $\mathbf{W}_j$, which represents the weight of the distance between the grid node and these matching points. In other words, the partial parallax information of the image is completely provided by inliers. APAP used SIFT and RANSAC respectively to extract features and eliminate mismatches. This combination can easily lead to insufficient or incorrect matching in multi-homography parallax scenes, resulting in obvious ghosting. Moreover, limited by the number of divided cells and the ability of rigid deformation, it can't handle the deformation in areas where the depth of field changes dramatically, which indicates that it is necessary to add post-processing step to ensure that the results are as aligned as possible. Besides, the perspective distortions also need to be corrected.

Based on the above analysis, the image stitching method proposed in this paper focuses on further eliminating the misalignments in general local warping approaches, which is mainly divided into four steps: image registration, local homography warping, warping post-processing and image fusion. The flow is shown in Fig. 1.

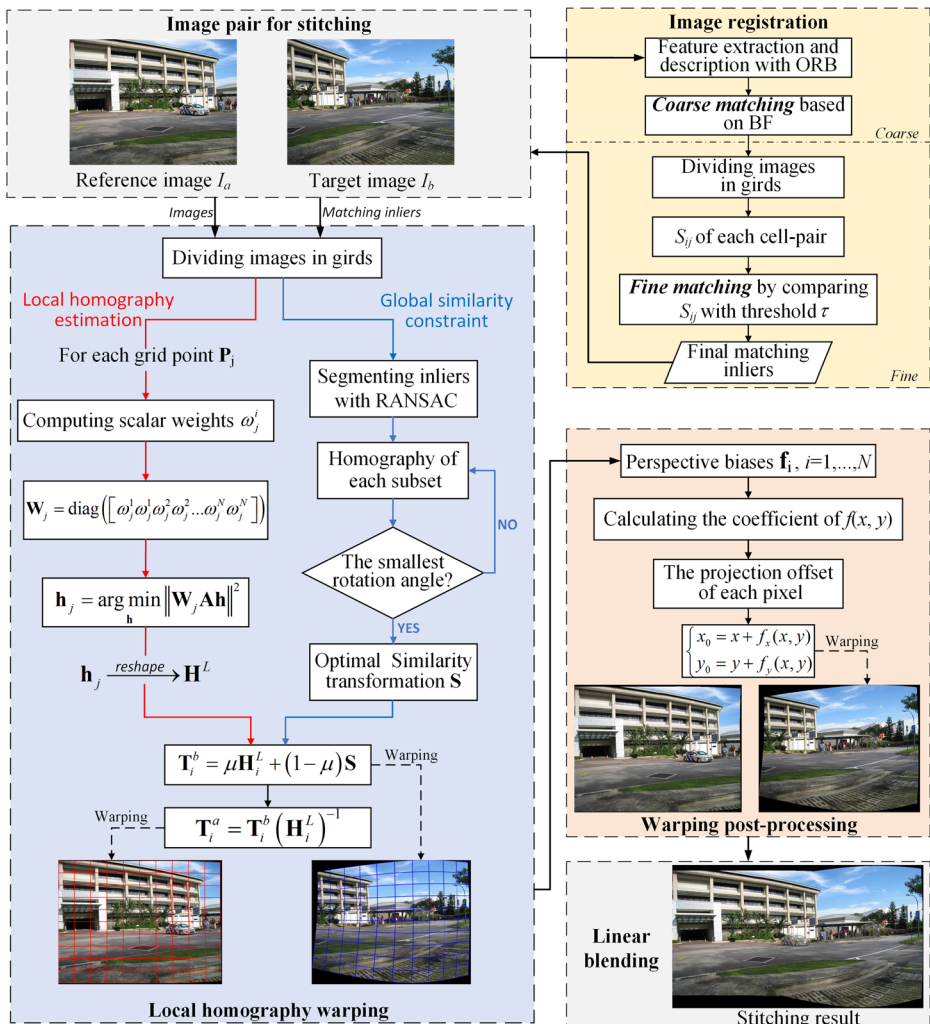


Fig. 1 Flow of the proposed method

First, ORB is used to extract a large number of feature points of the input images, and then pre-matching is performed through the BF algorithm to obtain coarse feature pairs. Next, the input images are divided into square grids and the GMS score of each cell-pair is calculated. The outliers are eliminated by comparing the GMS score with a set threshold, so as to obtain the final fine feature correspondences.

Next, the transformation model between the images is estimated according to the inliers, then the target image is mapped to the reference image coordinate system: The matching pairs are segmented into different subsets through RANSAC, the similarity transformation matrix corresponding to each subset is calculated and the one with the smallest rotation angle is selected. Then images are divided into grids, and the local homography of each grid is calculated according to the m-DLT algorithm. The similarity transformation and the local homography are weighted and fused to obtain the final transformation model, and the target image and the reference image are warped at the same time.

After local warping, further alignment is performed through the warping post-processing step based on TPS: For the projection biases that still exists after the local homography warping, a deformation function is constructed based on the corresponding relationship of inliers, so as to calculate the compensation offset of any misaligned pixel in the overlap area in both the horizontal and vertical directions. According to the offset, the ghosting area is locally adaptively adjusted to improve the alignment accuracy. Lastly, linear blending is used to achieve a satisfying seamless panorama result.

3.2.1 Image registration from coarse to fine

The first step in image stitching is to obtain accurate feature correspondences. Since GMS needs enough rough feature pairs for fine matching, the ORB algorithm is used for feature extracting from the two input images I_a and I_b . Then the rough point correspondences are obtained through Hamming distance-based brute force matching algorithm. $\{N_o, M_o\}$ is the number of the rough features in $\{I_a, I_b\}$ extracted and matched by ORB and BF, respectively.

Let $\chi = \{x_1, x_2, \dots, x_i, \dots, x_{N_o}\}$ be the nearest-neighbor matching set from I_a to I_b , χ can be classified as true or false by measuring the local support of each matching pair through GMS [2]. The region-pair $\{a, b\}$ is the regions of $\{I_a, I_b\}$ respectively, and each of them contains $\{n, m\}$ additional features. f_a represents one of the n supporting features in a , t denotes the correct matching probability. Given the same position $\{a, b\}$, let p_t represents the probability that the nearest neighbour of f_a is in region b . Similarly, p_f represents the probability that the nearest neighbour of f_a is in region b when $\{a, b\}$ view different locations. The GMS method assumes that the nearest neighbor match of f_a is in m possible positions. According to Bayes' rule, p_t and p_f are calculated as follows:

$$\begin{cases} p_t = t + (1 - t)\beta m/M_o \\ p_f = \beta(1 - t)(m/M_o) \end{cases} \quad (5)$$

where β is an adjustment parameter. Let S_i be the number of matches in the neighbourhood of x_i , the distribution of it can be approximated with a couple of binomial distribution:

$$\begin{cases} S_i \sim B(n, p_t), \text{ if } x_i \text{ is true} \\ S_i \sim B(n, p_f), \text{ if } x_i \text{ is false} \end{cases} \quad (6)$$

Based on the assumption of smoothness of motion over a large area proposed in [2], a more general method of score calculation is obtained as follows:

$$S_i = \sum_{k=1}^K |\chi_{a^k b^k}| - 1, \quad (7)$$

where K is the number of disjoint regions which match i predicts move together. $\chi_{a^k b^k} \subseteq \chi$ is the subset of matches that located on predicted region pairs $\{a^k, b^k\}$. Then the above assumption updates the binomial distribution in Eq. (6) to

$$\begin{cases} S_i \sim B(Kn, p_t), \text{ if } x_i \text{ is true} \\ S_i \sim B(Kn, p_f), \text{ if } x_i \text{ is false} \end{cases} \quad (8)$$

The key point of Eq. (8) is that both true and false matching have scores of supporting features in the region, but the distribution of S is quite different. That is, the probability distribution function of S may be bimodal, making the score of S an effective factor for distinguishing true and false matches. Thus, we focus on estimating the score S and threshold τ . To ensure speed and robustness, an image is divided into 20×20 non-repeating grids in [2]. But this will cause rectangular grid cells due to the difference in the length and width of the input image, resulting in uneven distribution of the coarse matching features in surrounding grids. In this paper, the number of divided grids is adaptively changed with the aspect ratio of input image, ensuring that the grids are as square as possible.

The S_{ij} of cell-pair $\{i, j\}$ is counted as shown below:

$$S_{ij} = \sum_{k=1}^{K=9} |\chi_{i^k j^k}|, \quad (9)$$

where $\chi_{i^k j^k}$ is the number of matches between $\{i^k, j^k\}$. $\{i^k, j^k\}$ are the nine grids centered on cell-pair $\{i, j\}$. The threshold τ for distinguishing whether cell-pairs are true matches is set as an empirical value, which is $\tau \approx \delta \sqrt{n}$. δ is an empirical parameter, which is usually set very large to ensure that most erroneous matches are reliably eliminated. The criterion of discrimination is summarized as: if $S_{ij} > \delta \sqrt{n_i}$, the matching pair x_i in cell-pair $\{i, j\}$ is true; otherwise x_i is false.

The refined N pair of matching points obtained after screening will be input as inliers into the subsequent warping.

3.2.2 Combination of global similarity transformation

Next, local homography warping is performed on the input images to achieve preliminary alignment. In order to provide natural-looking warping result, the global similarity transformation is integrated to address perspective distortions in non-overlapping areas.

The effectiveness of similarity transformation in reducing distortion has been proved by Chang et al. [7]. Different planes caused by lens movement can be approximated by similarity transformation, which can be applied to offset the perspective distortion [23]. In this method, different subsets of inliers are segmented by tuning the threshold of RANSAC to calculate corresponding similarity transformation. Then the transformation with the minimum rotation angle is chosen as the optimal

global similarity transformation. The final local warping is estimated by mixing the local homography and the global transformation as shown in Eq. (10):

$$\mathbf{T}_i^b = \mu \mathbf{H}_i^L + (1 - \mu) \mathbf{S}, \quad (10)$$

$$\mathbf{T}_i^a = \mathbf{T}_i^b (\mathbf{H}_i^L)^{-1} \quad (11)$$

Where the $\mathbf{H}_i^L$ is the i -th local homography estimated by Eq. (4), $\mathbf{S}$ is the optimal global similarity transformation, $\mathbf{T}_i^b$ is the final local warping transformation. μ is a weight coefficient between 0 and 1. The subscript b and a refer to the target image and reference image, respectively. Only applying the global similarity transform to the target image will cause misalignment in overlapping region, which needs the reference image to be compensated as (11) at the same time.

At last, the local warping results I'_a and I'_b are obtained by warping I_a and I_b with $\mathbf{T}^a$ and $\mathbf{T}^b$, respectively.

3.2.3 Warping post-processing based on TPS

For the projection biases that still appear after local warping, TPS is adopted to achieve further deformation to correct these deviations. The projective point of $\mathbf{p}$ in I_b can be denoted as $\mathbf{p}' = (x', y')^T$, which is obtained by local homography in Eq. (4). The projection bias can be denoted as $\mathbf{f} = \mathbf{p}' - \mathbf{q}$. Given specific correspondence $\{\mathbf{p}_i, \mathbf{q}_i\}$, projection bias is $\mathbf{f}_i = \mathbf{p}'_i - \mathbf{q}_i, i = 1, 2, \dots, N$. To facilitate subsequent calculations, $\mathbf{f}_i$ is divided into x -axis direction $f_x(x, y)$ and y -axis direction $f_y(x, y)$. Since their calculations are independent of each other, $f(x, y)$ is used to represent both in the following.

Define matrix:

$$\mathbf{L} = \begin{bmatrix} \mathbf{K} + \lambda \mathbf{E} & \mathbf{P} \\ \mathbf{P}^T & \mathbf{0} \end{bmatrix}, \mathbf{L} \in \mathbb{R}^{(N+3) \times (N+3)}, \quad (12)$$

where $\mathbf{K}_{i,j} = \phi(\|\mathbf{p}'_i - \mathbf{p}'_j\|), \phi(r) = r^2 \ln r$, the i -th row of $\mathbf{P}$ is $(1, x'_i, y'_i)$. λ is a coefficient which balances the influence of the smoothness of approximation and the accuracy. Let $\mathbf{F} = (f_1, \dots, f_N)$. The coefficients $\mathbf{e} = (\varepsilon_1, \dots, \varepsilon_N)$ and $\alpha_1, \alpha_2, \alpha_3$ are defined by the Eq. (13) as shown below:

$$\mathbf{L}^{-1}(\mathbf{F}|\mathbf{000})^T = (\mathbf{e}|\alpha_1\alpha_2\alpha_3)^T \quad (13)$$

According to the description of TPS [3], $f(x, y)$ can be expressed as follows:

$$f(x, y) = \alpha_1 + \alpha_2 x + \alpha_3 y + \sum_{i=1}^N \varepsilon_i \phi(\|\mathbf{p}'_i - (\mathbf{x}, \mathbf{y})\|_2) \quad (14)$$

Then, the projection bias of an arbitrary position in overlapping region can be estimated through Eq. (14).

In actual operation, the canvas is divided into $C3 \times C4$ cells, the projection offsets are first calculated only at the grid point. Then the offset of any other pixel can be linearly interpolated, which improves the speed and guarantees the accuracy at the same time. Based on the computed offset, I'_b can be further warped to $\hat{I}'_b$ to eliminate the misalignment with I'_a . More

specifically, let (x, y) be an arbitrary point in I'_b , (x_0, y_0) be its corresponding point in $\hat{I}'_b$. The transformation process is

$$\begin{cases} x_0 = x + f_x(x, y) \\ y_0 = y + f_y(x, y) \end{cases} \quad (15)$$

By calling inverse mapping using bilinear interpolation, the warping post-processing step is accomplished to further eliminate projection biases. It can be seen from (14) that the calculation accuracy of pixel point offsets is related to the quality of anchor points. In other words, the higher the number and accuracy of inliers obtained by image matching, the stronger the TPS alignment ability. This is also one of the reasons why ORB-GMS matching algorithm is adopted in this paper.

At last, linear blending is used to fuse overlapping areas, making the seams almost invisible.

4 Experiments and analysis

To evaluate the capability of this method, a set of experiments are performed in this section. These include: 1) Qualitative and quantitative comparison of alignment quality with other representative methods; 2) Testing the reduction of perspective distortion after introducing global similarity constraint; 3) Evaluating the alignment quality by stages to determine the contribution of registration strategy and the warping post-processing; 4) Comparison of computational efficiency, respectively. The compared methods are global homography, APAP [32], SPHP [7], AANAP [23], Fast and Robust Seam estimation (FARSE) [17] and Deep Image Stitching (DIS) [26]. Among them, global homography represents the traditional methods with global transformation model, APAP is the representative of spatially-varying warping model. SPHP and AANAP are representative warping algorithms with perspective distortion elimination. FARSE is a representative seam cutting algorithm in recent years. DIS is a deep learning based image stitching method. All test algorithms run on the same hardware platform, with intel i5–8400 2.8 GHz CPU and 16 GB RAM. All these databases used in this section are collected from public datasets [7, 8, 12, 20, 21, 23, 24], which have been widely used in image stitching.

The parameter setting of these comparison methods is consistent with the corresponding paper. If a range of values is provided, the parameter with the best effect is selected. For our method, the initial number of ORB points extracted is set to 10,000, δ is set to 6, $C1 = C2 = 200$, $C3 = C4 = 100$, $\gamma = 0.1$, λ is 0.1% of the size of input image.

4.1 Qualitative comparisons

4.1.1 Qualitative comparison on alignment quality

As shown in Fig. 2, the *worktable* [20] database is a challenging situation. The results of each stitching method are arranged in rows, with obvious misalignments highlighted and magnified. As a basis for comparison, the result of global homography is shown in Fig. 2a. Since only the global homography of two images is calculated, misalignment due to projection biases can be seen clearly in these extracted magnified areas. Thanks to the application of local homography

warping, compared to Fig. 2a, the stitching quality of APAP has been improved to a certain extent, but the misalignments still exist, which is shown in Fig. 2b (I–IV). For SPHP, due to the lack of a more precise alignment strategy, the ghosting in Fig. 2c is the most severe among several methods. Similar to APAP, the result of AANAP also contains some slight ghosting, as shown in Fig. 2d. This is because AANAP only improves on perspective distortion, but its alignment strategy is actually consistent with APAP. FARSE only roughly aligns the images with global homography, the limited alignment ability makes it hardly to find a stitching seam in the overlap region that completely bypasses the ghost area in large parallax scenes, resulting in misalignment, as shown in Fig. 2e(I) and (IV). With the help of higher-quality registration strategy based on GMS and TPS-based warping post-processing step, our results successfully eliminated the remaining projection biases in (b) (d), and achieved precise alignment in the four highlighted areas (Fig. 2f (I–IV)).

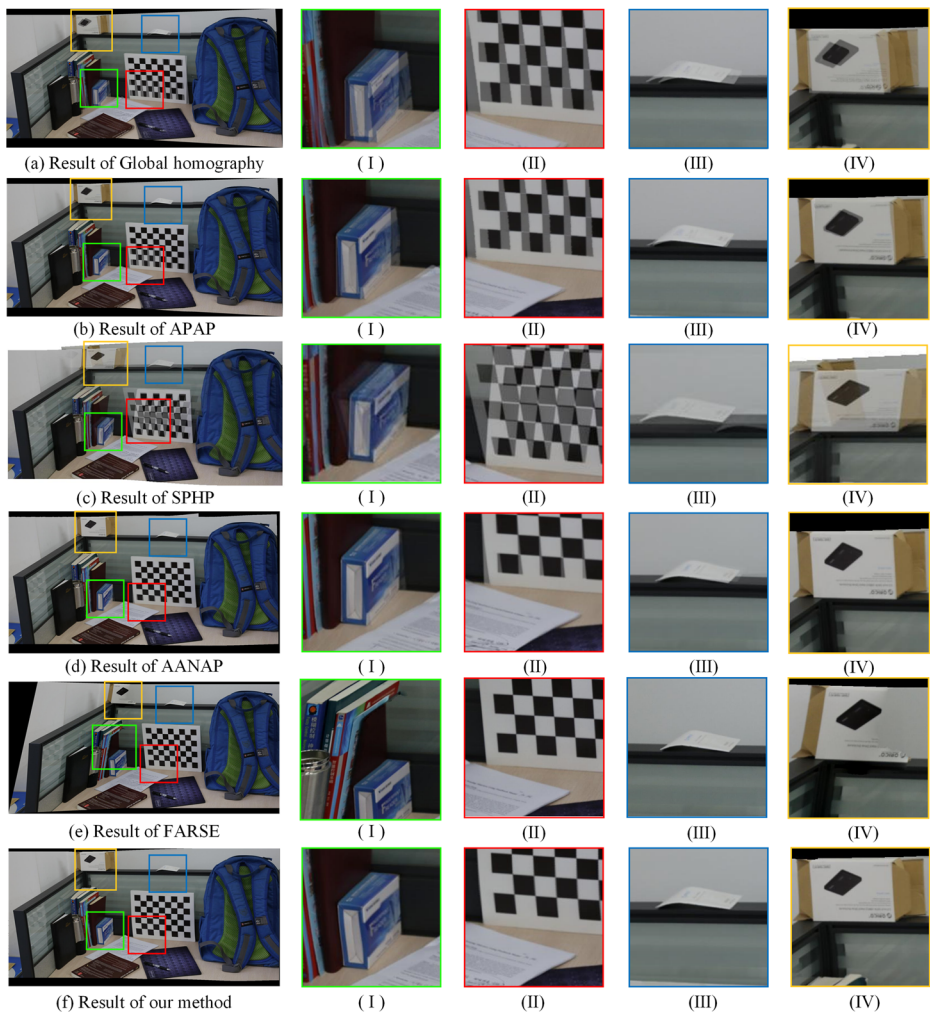


Fig. 2 Comparison of alignment quality on *worktable* scene. **a** Result of global homography. **b** Result of APAP. **c** Result of SPHP. **d** Result of AANAP. **e** Result of FARSE. **f** Result of our method. The green, red, blue and yellow boxes in each of the results are enlarged and are named (I), (II), (III) and (IV) respectively

4.1.2 Qualitative comparison on perspective distortion reduction

In this section, a case with both parallax and perspective distortion problems was used for comparison. The results comparison of *temple* database [12] are shown in Fig. 3. Among them, the red and green boxes highlight the representative areas in overlapping region, and the blue box highlights the perspective distortion in non-overlapping region.

There is obvious ghosting in Fig. 3a (I-II). And the global homography is directly extrapolated to non-overlapping areas, resulting in the severe tilt of the two buildings and their adjacent shopping mall in Fig. 3a (III). In Fig. 3b, local warping helps APAP eliminate the misalignment in the green box. However, the misalignment in the red box has not been completely corrected, and the perspective distortion in Fig. 3b(III) has not been improved. SPHP and AANAP focus on eliminating these perspective distortions, so the inclination of the buildings in Fig. 3c(III) and Fig. 3d(III) has been significantly reduced. However, the ghosting of the roof and tree in Fig. 3c(II) in SPHP is serious. There is less ghosting in AANAP, but there are still some misalignments in Fig. 3c(II) such as tree branches and lamp posts. FARSE has a severe misalignment in in Fig. 3e (II), and the perspective distortion in the upper left area of the stitching result is serious, which is highlighted by yellow box. Figure 3f shows the results of our method, the buildings in (III) look natural, and the areas in (I-II) are accurately aligned at the same time.

By adding grids to input images, Fig. 4 can show the effect of adopting global similarity constraints more intuitively. Figure 4a shows the warping of the input images by APAP. APAP aims to adjust the local deformation while retaining the global projection, so the global homography is directly extrapolated to the non-overlapping region, resulting in the stitching result affected by the perspective distortion. As a contrast, the objects in the non-overlapping area of Fig. 4b can maintain their original appearance after integrating the global similarity constraint. This comparison shows that our method can not only eliminate projection biases, but also largely mitigate perspective distortion in non-overlapping regions, providing satisfactory stitching results.

4.1.3 Qualitative comparison with learning based method

The representative of deep learning stitching methods, DIS, has its particularity: It cannot handle input images of arbitrary resolution (the resolution is only 128*128), and it only performs well on synthetic image sets without parallax. Therefore, for the fairness of comparison, the stitching result of DIS is separately compared with our method, as shown in the Fig. 5. It can be seen that DIS has good alignment ability in parallax scenes in the first two columns, but obvious blur occurs at the edges of the results. When the perspective distortion of the image is severe (the last three columns), the non-overlapping area will produce serious distortion, the stitching even fails in the fifth scene. In contrast, the proposed method achieves good alignment and maintains natural look in all parallax scenes.

Through experimental comparison, the current deep image stitching methods are insufficient to meet the needs of practical applications. There are two main reasons: First, the networks adopted by deep methods usually have strict limitations on the size of the input image, resulting in blur or deformation in stitching results. Second, they are all supervised methods, but since the labels are not available in the actual scenes, there is no real training dataset. Therefore, traditional methods are still the mainstream in the field of image stitching.

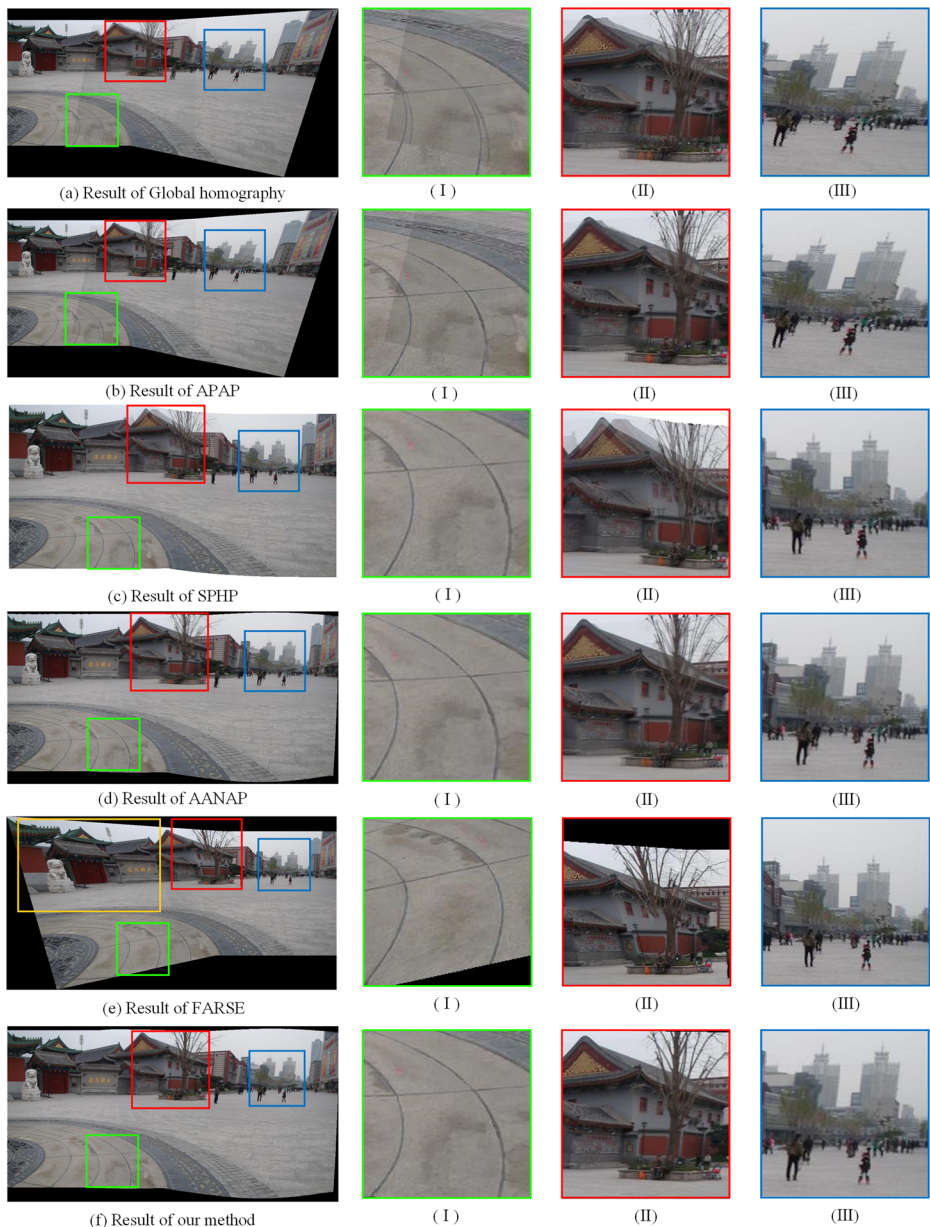


Fig. 3 Comparison of alignment quality and perspective distortion reduction on *temple* scene. **a** Result of global homography. **b** Result of APAP. **c** Result of SPHP. **d** Result of AANAP. **e** Result of FARSE. **f** Result of our method

4.2 Quantitative analysis

Commonly used image quality assessment indicators, such as PSNR and SSIM, focus on global distortion categories of images such as noise and blur, and cannot precisely measure the alignment accuracy of the stitched results [31]. Additionally, there is no reference image to

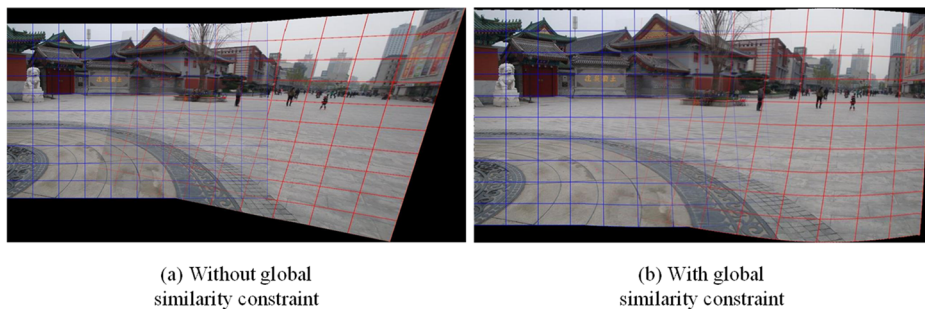


Fig. 4 Warping of global similarity constraint in reducing perspective distortions

assist in the assessment because of the unavailability of the panorama in actual scene. Therefore, in this section, root mean square error (RMSE) [19] is adopted to measure the aligning quality of different methods quantitatively, which is defined as

$$RMSE(I, I') = \sqrt{\frac{1}{N_p} \sum_{\pi} (I - NCC(\mathbf{p}_a, \mathbf{p}_b))^2} \quad (16)$$

where π denotes the overlapping region, the number of pixels in π is N_p . $\mathbf{p}_a$ and $\mathbf{p}_b$ respectively represent pixels in I_a and I_b , and they are in the same position in the overlapping area. NCC is the normalized cross correlation over a neighborhood of 3×3 window. RMSE shows the accuracy of alignment of the two warped images in overlapping area. The lower the RMSE value, the higher the alignment quality.

All these databases used for testing are collected from public datasets, including a variety of cases such as small or large parallax, weakly-textured regions and repetitively-textured regions. FARSE and DIS adopted the seam cutting and deep learning optimization methods to eliminate artifacts respectively, thus RMSE was not suitable to be used as an evaluation index for them. The computed scores of the remaining five approaches on different databases are shown in Table 1.

For the first nine cases with parallax, thanks to the integration of local warping, the alignment quality of the latter four methods is higher than the global homography method. Since our method uses TPS for warping post-processing, the deviations that are difficult to align in other methods are effectively eliminated, so the best alignment is achieved in most cases. The last five cases of images have weakly-textured regions and repetitively-textured regions, which brings challenges for feature extraction and matching. Other methods adopt the

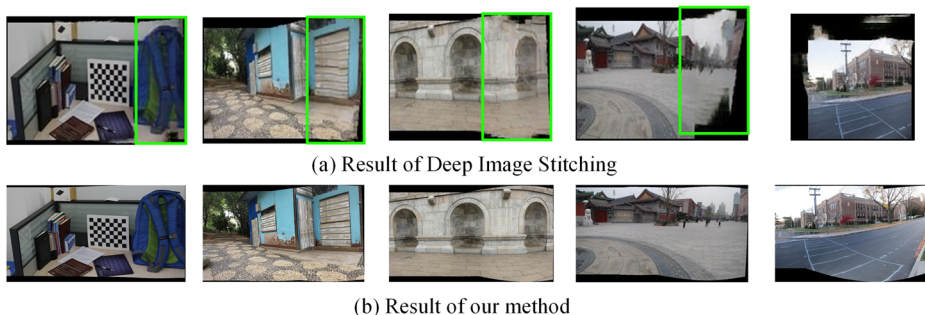


Fig. 5 The comparison of the stitching results between DIS and our method

Table 1 RMSE of each image pair for five comparison methods

Database	Global	APAP	SPHP	AANAP	Ours
<i>temple</i>	8.823	7.905	6.503	6.732	5.457
<i>garden</i>	27.668	25.296	19.584	19.023	14.586
<i>chessgirl</i>	7.523	6.987	5.253	5.993	5.916
<i>Cathedral</i>	7.854	6.069	14.663	6.962	6.273
<i>cabin</i>	14.484	13.209	12.929	8.823	6.936
<i>railtracks</i>	22.389	28.101	16.040	10.073	6.503
<i>theater</i>	6.299	5.304	9.869	4.157	3.596
<i>carpark</i>	16.550	13.235	10.991	9.894	6.936
<i>seagull-24</i>	8.976	8.109	5.483	6.758	6.222
<i>wall</i>	8.645	9.180	5.355	4.769	4.106
<i>building</i>	19.176	13.796	10.736	9.206	6.707
<i>tower</i>	24.761	19.431	9.920	13.082	7.115
<i>desk</i>	5.636	5.279	5.712	5.253	4.667
<i>worktable</i>	11.909	6.528	17.264	8.186	5.024

registration combination of SIFT+RANSAC. In comparison, our ORB+GMS coarse-to-fine registration method can obtain more number and more accurate matches, thereby improving the alignment accuracy. Therefore, the score of our method is lower, which means that the stitching result is better.

4.3 Alignment quality evaluation

In this method, there are three main aspects that contribute to the improvement of the stitching quality: 1) the matching from coarse to fine, 2) the warping post-processing based on TPS, and 3) the global similarity constraints. Since the contribution of global similarity has been demonstrated in Section 4.1.2, this section focusses on the alignment capability of the first two aspects. AANAP is used as the basic method, and the alignment quality of the results is evaluated qualitatively and quantitatively in stages.

Figure 6 shows the improvement in the alignment quality of the two aspects in the proposed method on the *worktable* database. Figure 6b shows the stitching result after only GMS registration is introduced. It can be seen that compared with (a), the ghosting in (b) is significantly reduced, but there is still room for improvement. As shown in (c), after introducing the warping post-processing, the misalignments in these three boxes are completely eliminated, providing a pleasing stitching result. The *tower* database in Fig. 7 is representative of the low-texture situation. The tower body has been aligned at the beginning, thus we focus on the result of the stitching of the ground tiles (highlighted by red boxes) at different stages. Through comparison, it is obvious that the alignment quality has gradually improved. This is because the ORB-GMS matching combination is not restricted by the global homography, so a sufficient number of matches can be retained in the ground area, providing more data support for calculating the local homography of this region and the subsequent TPS deformation.

The staged RMSE results for representative databases are shown in Table 2. After the introduction of the coarse-to-fine matching, the RMSE value is reduced, and the reduction in low texture parallax conditions (the latter three) is relatively larger. This demonstrates that the registration strategy of ORB+GMS can describe the parallax information more comprehensively and provide more feature correspondence. TPS is an effective alignment tool, therefore after warping post-processing step, the RMSE value in all cases is significantly reduced. The

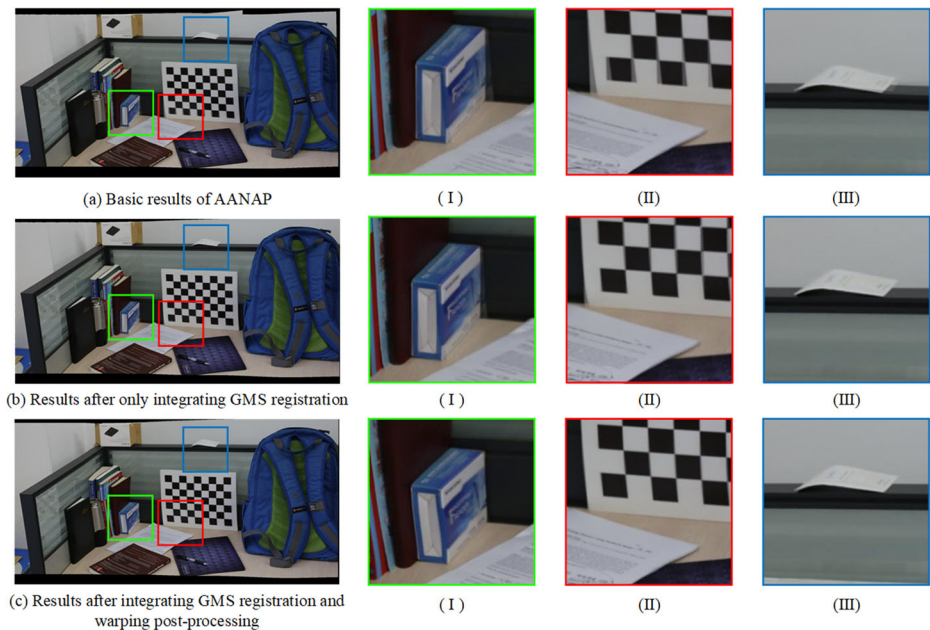


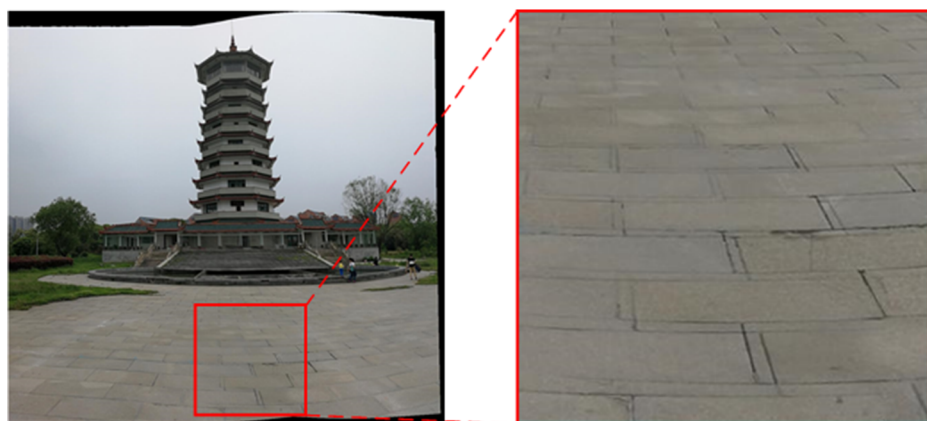
Fig. 6 Comparison by stages on *worktable* scene. **a** Basic results of AANAP. **b** Results after only integrating GMS registration. **c** Results after integrating GMS registration and warping post-processing. The green, red and blue boxes in each of the results are enlarged and are named (I), (II), and (III) respectively

results of qualitative and quantitative experiments show the effectiveness of the optimizations proposed in this method for improving the alignment of stitching.

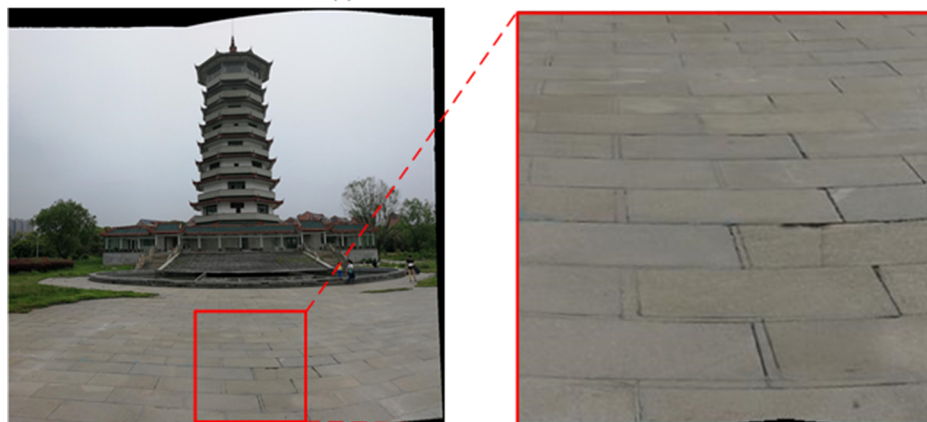
4.4 Comparison of computational efficiency

The computational efficiency of each representative method is counted and compared in this section, including APAP, SPHP, AANAP, FARSE and DIS. Since the input resolution of DIS is fixed and the software platform (Tensorflow 1.13.1) is different from other methods, the elapsed time of DIS is for reference only.

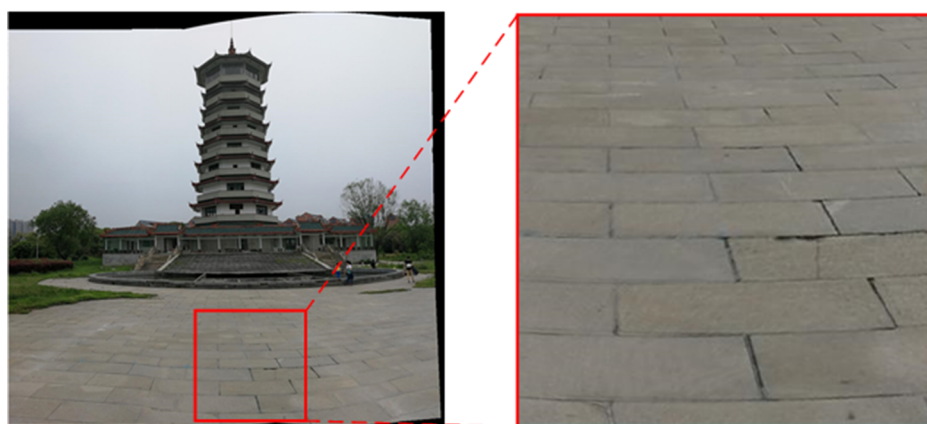
The elapsed time comparison is shown in Table 3. The neural network used by DIS has strong generalization ability and compresses all the input images to the same size, so the elapsed time in different cases is almost the same. DIS has the highest computational efficiency, but its stitching quality is also the worst at the same time. FARSE only roughly aligns the images and eliminates artifacts through optimal seam, which is the most efficient in traditional methods, but sacrifices the stitching quality in large parallax scenes. Among several types of spatially-varying warping methods, compared with AANAP, APAP lacks constraints on global similarity, and SPHP lacks local homography calculations. Therefore, in most scenes, APAP is the fastest, SPHP is the second, AANAP is the slowest, and the order of alignment accuracy is reversed. The method in this paper aims to align the images as accurately as possible, so the TPS-based projection bias elimination step is added. Although ORB-GMS takes less time than SIFT-RANSAC, the final elapsed time is still slower than AANAP.



(a) Basic results of AANAP



(b) Results after only integrating GMS registration



(c) Results after integrating GMS registration and warping post-processing

Fig. 7 Comparison by stages on *tower* scene. **a** Basic results of AANAP. **b** Results after only integrating GMS registration. **c** Results after integrating GMS registration and warping post-processing

Table 2 RMSE for our methods at different stages

Database	Stages		
	Original AANAP	+GMS registration	+ warping post-processing
<i>temple</i>	6.732	6.197	5.483
<i>garden</i>	19.023	18.131	14.688
<i>cabin carpark</i>	8.823	8.466	6.477
	9.894	8.798	6.962
<i>worktable</i>	8.186	7.089	5.075
<i>tower</i>	13.082	9.818	7.166
<i>building</i>	9.206	8.007	6.605

The comparison shows that the method in this paper sacrifices computational efficiency to obtain the highest possible alignment accuracy, therefore is suitable for applications that require high quality stitching results but low real-time requirements.

5 Conclusion

A misalignment-eliminated warping image stitching method is proposed in this paper. In view of these misalignments that still exist in the results of general local warping methods, we have made improvements in two aspects: For one thing, a registration strategy from coarse to fine is adopted in image registration step to provide sufficient accurate inliers. For another, a post-processing step is added after local warping to construct an elastic warping model to further correct the projection biases. Additionally, global similarity constraint is integrated to ensure that the stitching result will not have obvious perspective distortion. The results of experiments show that the proposed method can align the images as accurately as possible to obtain higher-quality panoramic stitching results. But this method does not perform well in the cases of severe occlusion, and the calculation time is longer. Future work will focus on solving occlusion problems and improving computational efficiency.

Table 3 Comparison of the elapsed time of different methods

Database	Image size	The elapsed time (s)					
		APAP	SPHP	AANAP	FARSE	DIS	Ours
<i>temple</i>	730×487	4.96	8.28	6.28	3.41	4.69	8.97
<i>garden</i>	1440×1920	20.96	18.81	24.67	14.35	4.70	27.98
<i>chessgirl</i>	1008×755	6.01	10.11	7.17	4.52	4.68	8.31
<i>Cathedral</i>	2448×3264	65.47	57.06	73.79	48.28	4.69	82.53
<i>cabin</i>	1280×960	11.75	15.19	14.39	7.11	4.70	16.77
<i>railtracks</i>	2000×1500	29.53	33.75	35.94	23.19	4.69	41.87
<i>theater</i>	2160×1440	32.57	28.04	37.09	26.12	4.69	43.79
<i>carpark</i>	653×490	3.28	8.40	5.92	2.54	4.69	6.43
<i>seagull-24</i>	1000×669	8.98	8.11	12.48	6.76	4.68	15.22
<i>wall</i>	1280×960	8.07	15.41	13.19	8.67	4.69	17.17
<i>building</i>	720×960	6.12	10.13	9.03	4.75	4.69	11.36
<i>tower</i>	960×1280	7.19	13.15	10.18	6.62	4.72	13.73
<i>desk</i>	500×375	1.58	7.55	3.51	2.86	4.70	4.89
<i>worktable</i>	2160×1440	25.44	25.72	30.44	17.38	4.71	34.08

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Declarations

Conflict of interest The authors declare no conflict of interest.

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